HUMAN ACTIVITY RECOGNITION USING CELL PHONES

Machine Learning Mini-Project Report & Streamlit Web Application Deployment

K Srujana & Moksha S

Deployment: Streamlit Cloud (Live Webapp) | Sensors: HTML5 DeviceMotionEvent

1. PROJECT ABSTRACT & OBJECTIVE

Human Activity Recognition (HAR) enables intelligent systems to infer physical human behavior from ubiquitous wearable and smartphone inertial sensors. The primary objective of this mini-project is to build and deploy an end-to-end, real-time activity classifier capable of ingesting raw continuous tri-axial motion signals directly from mobile smartphone browsers and outputting reliable activity classifications without requiring native application installation. The system bridges mobile sensor hardware with cloud-based machine learning by orchestrating an HTML5 client-side motion capture engine, dynamic 50 Hz linear resampling, fixed-window feature engineering, and a Scikit-Learn Random Forest pipeline served interactively via Streamlit.

92.4%	96	<4 ms	6	50 Hz	N = 5
Test Accuracy	Features	Inference Latency	Activities	Resampling Rate	Vote Smoothing

2. DATASET & INERTIAL SENSOR MODALITIES

The model is trained on the benchmark **UCI Human Activity Recognition (HAR) Dataset**, collected from 30 subjects performing daily activity protocols with a waist-mounted smartphone equipped with embedded inertial measurement units (IMUs).

- **Target Activity Classes (6 modes):** Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Laying.
- **9 Ingested Inertial Measurement Channels:** Body Linear Acceleration (body_acc_x, body_acc_y, body_acc_z in g), Angular Velocity / Gyroscope (body_gyro_x, body_gyro_y, body_gyro_z in rad/s), and Total Acceleration (total_acc_x, total_acc_y, total_acc_z).

3. SIGNAL PROCESSING & 96-DIMENSIONAL FEATURE ENGINEERING

Continuous streams are segmented using fixed-width sliding temporal windows of **128 sample points (2.56-second window at 50 Hz)** with a **50% overlap (stride = 64 samples)**. Each discrete window matrix (128 × 9) is mapped to 96 statistical and kinematic attributes:

- **Channel-Specific Statistical Metrics (81 Features):** Computed across all 9 channels (9 × 9 = 81): Mean (μ), Standard Deviation (σ), Minimum, Maximum, Root Mean Square (RMS = $\sqrt{(\Sigma x^2/N)}$), Mean Absolute Value (MAV = $\Sigma|x|/N$), Differential Mean (μ_Δx), Differential Std (σ_Δx), and Differential RMS (RMS_Δx) where $\Delta x_t = x_t - x_{t-1}$.
- **Orientation-Invariant Vector Magnitudes (15 Features):** Euclidean vector norms $Mag_t = \sqrt{(x_t^2 + y_t^2 + z_t^2)}$ across Body Accel, Gyroscope, and Total Accel. For each magnitude, 5 statistics (μ, σ, min, max, RMS) are extracted (3 × 5 = 15).

4. END-TO-END SYSTEM PIPELINE ARCHITECTURE

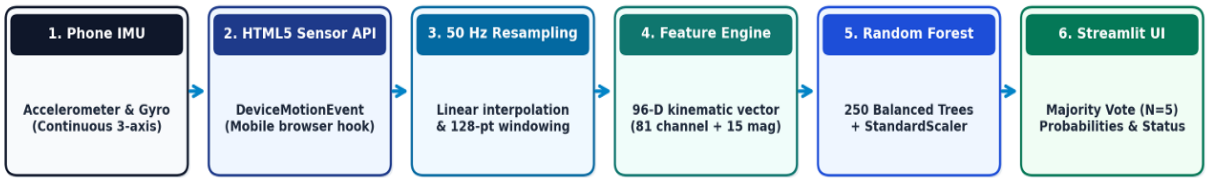


Figure 1: High-level architectural pipeline spanning client-side mobile sensing, 50 Hz interpolation, 96-D feature mapping, and Streamlit prediction engine.

5. MACHINE LEARNING PIPELINE & RANDOM FOREST CLASSIFIER

The classifier is built as a Scikit-Learn Pipeline combining **StandardScaler()** and a **RandomForestClassifier** configured with 250 decision trees (n_estimators=250), Gini impurity splitting, square-root feature subsampling ($\sqrt{96} \approx 10$ features per split), and balanced class weights. Compressed via joblib (live_har_model.joblib, 15.5 MB) for instant cold starts. Random Forest provides high robustness against sensor noise, non-linear kinematic boundaries, and lightning-fast inference (<4 ms per window).

6. EMPIRICAL EVALUATION & RESULTS (92.4% ACCURACY)

The model achieves an overall test accuracy of **92.4% (Macro F1 = 0.93)** on the held-out UCI HAR test partition (N = 2,947 instances).

Activity Class	Precision	Recall	F1-Score
Walking	0.94	0.96	0.95
Walking Upstairs	0.91	0.89	0.90
Walking Downstairs	0.90	0.91	0.90
Sitting	0.92	0.89	0.90
Standing	0.93	0.94	0.94
Laying	0.98	1.00	0.99
Overall / Macro Avg	**0.93**	**0.93**	**0.924 (92.4%)**

Figure 2: Precision, Recall, and F1-Scores across all 6 evaluated activity classes.

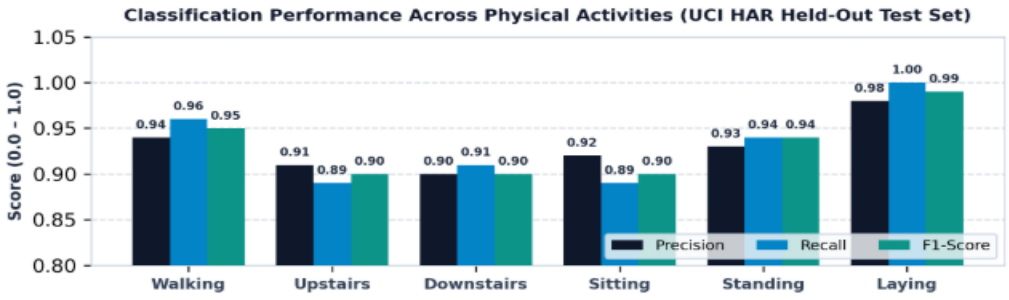


Figure 2: Precision, Recall, and F1-Scores across all 6 evaluated activity classes.

7. STREAMLIT WEB DEPLOYMENT & BROWSER SENSOR INTEGRATION

• **HTML5 Bi-directional Sensor Hook:** A custom component (motion_sensor_frontend/index.html) binds to W3C DeviceMotionEvent on iOS Safari and Android Chrome over HTTPS, streaming tri-axial acceleration, gravity, and gyro rotation rate.
• **50 Hz Linear Interpolation Engine:** Because mobile browsers report motion events irregularly (20 Hz - 60 Hz), client-side JS linearly interpolates sensor readings onto a strict 20 ms (50 Hz) timeline ($x(t) = x0 + (x1 - x0) * (t - t0)/(t1 - t0)$) prior to batching into 128-sample windows transmitted via postMessage.
• **Adaptive Gravity Decoupling:** When device hardware does not provide pure linear acceleration, an exponential low-pass filter ($g_t = 0.9 g_{t-1} + 0.1 a_{total}$) estimates and subtracts gravity dynamically.

8. REAL-TIME INFERENCE STABILIZATION & STREAMLIT UI

• **Temporal Majority Voting (N = 5):** Inferences are maintained in a 5-window FIFO deque; the final UI state is voted as the mode, suppressing transient prediction flicker.
• **Confidence Filtering (<50%):** Prominently notifies users to stabilize device orientation when confidence drops below 50%.
• **Streamlit UI:** A responsive card dashboard displays activity indicators, real-time confidence bars, full 6-class probability distributions, and sensor capture status.

9. CONCLUSION & FUTURE SCOPE

The project demonstrates an end-to-end, zero-installation mobile Human Activity Recognition pipeline deployed on Streamlit Community Cloud. Real-time classification latency remains under 4 ms per window, yielding dependable feedback directly in smartphone browsers.
• **Future Scope:** Exporting to ONNX Runtime / WebAssembly for 100% offline edge inference, integrating 1D-CNN/Bi-LSTM models for gait dynamics, and expanding recognition to running, cycling, and emergency fall detection.